

Week 2: Product and Category Management System

Scenario:

The e-commerce company wants to maintain information about products and their categories. Each product belongs to a specific category, and the system should manage product details such as name, price, and stock availability.

1. Design Product and Category Tables

Category Table:

```
SQL> CREATE TABLE Item_Category(Cat_ID VARCHAR2(7),
  2  Category_Title VARCHAR2(20),
  3  Category_Desc VARCHAR2(50));

Table created.
```

Product Table:

```
SQL> CREATE TABLE Product_Catalog(Prod_ID VARCHAR2(7),
  2  Prod_Name VARCHAR2(20),
  3  Cat_ID VARCHAR2(7),
  4  Item_Price NUMBER(9,2),
  5  Available_Qty Number);

Table created.
```

The two tables for Category and Product have been created.

Now, we have to find the relationship between the tables and apply constraints to the tables.

2. Define Relationships and Constraints

Relationships:

One Category Can have many Products.

The relationship between Category and Product is One to many (1-> *)

We found that, what is their relationship. Now we have to discuss that which is the key that is going to make this relationship.

Before that, we need to understand about KEY Constraints.

PRIMARY KEY: The Column(s) or Field(s) that uniquely identifies a row/ tuple

In these two tables, which field can be a Primary Key?

In Category Table, Cat_ID is a unique Value that helps to identify each category. Likewise, in Product Table Prod_ID is a unique Column.

FOREIGN KEY: The Key which links two tables.

Therefore, Cat_ID acts as Foreign Key in the Product Table.

We can explore some other constraints like NOT NULL, CHECK and UNIQUE

NOT NULL: Ensures that the Column cannot have a empty (null) value.

For Example, a Product name or Category name cannot be empty. All category or product must have a name.

CHECK: Validates the values in a column satisfies the condition.

For Example, To check the Price of a Product must always be a positive number.

UNIQUE: Ensures that there is no repeated value for a column.

For Example, If we mention UNIQUE Constraint to Category Name, no two category can have same name.

Defining Relationship and Creating Constraints for Product and Category:

Primary Key:

```
SQL> ALTER TABLE Item_Category
  2  ADD CONSTRAINT PK_Item_Category PRIMARY KEY (Cat_ID);

Table altered.

SQL> ALTER TABLE Product_Catalog
  2  ADD CONSTRAINT PK_Product_Catalog PRIMARY KEY (Prod_ID);

Table altered.
```

Creating Relationship using FOREIGN KEY:

```
SQL> ALTER TABLE Product_Catalog
  2  ADD CONSTRAINT FK_Prod_Category FOREIGN KEY (Cat_ID)
  3  REFERENCES Item_Category(Cat_ID);

Table altered.
```

Adding Constraints:

NOT NULL:

```
SQL> ALTER TABLE Item_Category
  2  MODIFY Category_Title VARCHAR(100) NOT NULL;

Table altered.

SQL> ALTER TABLE Product_Catalog
  2  MODIFY Prod_Name VARCHAR(150) NOT NULL;

Table altered.
```

CHECK:

```
SQL> ALTER TABLE Product_Catalog
  2 ADD CONSTRAINT CHK_Item_Price CHECK (Item_Price > 0);

Table altered.

SQL> ALTER TABLE Product_Catalog
  2 ADD CONSTRAINT CHK_Available_Qty CHECK (Available_Qty >= 0);

Table altered.
```

UNIQUE:

```
SQL> ALTER TABLE Item_Category
  2 ADD CONSTRAINT UQ_Category_Title UNIQUE (Category_Title);

Table altered.
```

3. Insert Values:

```
SQL> INSERT INTO Item_Category
  2 VALUES(
  3 '&Cat_ID',
  4 '&Category_Title',
  5 '&Category_Desc');
Enter value for cat_id: CAT-101
old 3: '&Cat_ID',
new 3: 'CAT-101',
Enter value for category_title: Computing & Peripherals
old 4: '&Category_Title',
new 4: 'Computing & Peripherals',
Enter value for category_desc: Workstations, displays, and computer hardware
old 5: '&Category_Desc')
new 5: 'Workstations, displays, and computer hardware')

1 row created.

SQL> /
Enter value for cat_id: CAT-102
old 3: '&Cat_ID',
new 3: 'CAT-102',
Enter value for category_title: Activewear & Footwear
old 4: '&Category_Title',
new 4: 'Activewear & Footwear',
Enter value for category_desc: Performance sportswear, shoes, and outdoor gear
old 5: '&Category_Desc')
new 5: 'Performance sportswear, shoes, and outdoor gear')

1 row created.
```

Product Table:

```
SQL> INSERT INTO Product_Catalog
  2 VALUES(
  3 '&Prod_ID',
  4 '&Prod_Name',
  5 '&Cat_ID',
  6 &Item_Price,
  7 &Available_Qty);
Enter value for prod_id: PRD-101
old 3: '&Prod_ID',
new 3: 'PRD-101',
Enter value for prod_name: Ultrabook Pro 15
old 4: '&Prod_Name',
new 4: 'Ultrabook Pro 15',
Enter value for cat_id: CAT-101
old 5: '&Cat_ID',
new 5: 'CAT-101',
Enter value for item_price: 74500
old 6: &Item_Price,
new 6: 74500,
Enter value for available_qty: 4
old 7: &Available_Qty)
new 7: 4)

1 row created.
```

Like this insert few more rows.

After Insertion Category Table:

```
SQL> SELECT * FROM Item_Category;

CAT_ID      CATEGORY_TITLE
-----
CATEGORY_DESC
-----
CAT-101      Computing & Peripherals
Workstations, displays, and computer hardware

CAT-102      Activewear & Footwear
Performance sportswear, shoes, and outdoor gear

CAT-103      Smart Home Living
Sensors, automated lighting, and appliances

CAT_ID      CATEGORY_TITLE
-----
CATEGORY_DESC
-----
CAT-104      Publications & Media
Technical reference manuals and literature
```

Product Table

```
SQL> SELECT * FROM Product_Catalog;
```

PROD_ID	PROD_NAME	CAT_ID	ITEM_PRICE	AVAILABLE_QTY
PRD-101	Ultrabook Pro 15	CAT-101	74500	4
PRD-102	Mechanical Keyboard	CAT-101	4800	9
PRD-201	Trail Running Shoes	CAT-102	3600	6
PRD-202	Thermal Windbreaker	CAT-102	1850	12
PRD-301	Smart Air Purifier	CAT-103	16500	5
PRD-401	Clean Code Handbook	CAT-104	899	7
PRD-402	Design Patterns Guide	CAT-104	650	3

7 rows selected.

CRUD Operations:

Insert Record:

```
SQL> INSERT INTO Product_Catalog (Prod_ID, Prod_Name, Cat_ID, Item_Price, Available_Qty)
  2  VALUES ('PRD-302', 'Robotic Vacuum Cleaner', 'CAT-103', 24000, 8);

1 row created.
```

Update Price:

```
SQL> UPDATE Product_Catalog SET Item_Price = 69900 WHERE Prod_Name = 'Ultrabook Pro 15';

1 row updated.
```

Delete a Product:

```
SQL> DELETE FROM Product_Catalog WHERE Prod_Name = 'Design Patterns Guide';

1 row deleted.
```

5. Generate Category-wise Product Reports

Write SQL queries to:

- Display all products under each category.

```
SQL> SELECT c.Category_Title, p.Prod_Name, p.Item_Price, p.Available_Qty
  2  FROM Product_Catalog p JOIN Item_Category c ON p.Cat_ID = c.Cat_ID;
```

CATEGORY_TITLE	PROD_NAME	ITEM_PRICE	AVAILABLE_QTY
Computing & Peripherals	Ultrabook Pro 15	69900	4
Computing & Peripherals	Mechanical Keyboard	4800	9
Activewear & Footwear	Trail Running Shoes	3600	6
Activewear & Footwear	Thermal Windbreaker	1850	12
Smart Home Living	Smart Air Purifier	16500	5
Smart Home Living	Robotic Vacuum Cleaner	24000	8
Publications & Media	Clean Code Handbook	899	7

7 rows selected.

- Count number of products in each category.

```
SQL> SELECT c.Category_Title, COUNT(p.Prod_ID) AS Product_Count
  2  FROM Product_Catalog p JOIN Item_Category c ON p.Cat_ID = c.Cat_ID
  3  GROUP BY c.Category_Title;
```

CATEGORY_TITLE	PRODUCT_COUNT
Computing & Peripherals	2
Activewear & Footwear	2
Smart Home Living	2
Publications & Media	1

- Find highest-priced product in each category.

```
SQL> SELECT c.Category_Title, p.Prod_Name, p.Item_Price AS Highest_Price
  2  FROM Product_Catalog p
  3  JOIN Item_Category c ON p.Cat_ID = c.Cat_ID
  4  WHERE p.Item_Price = (
  5      SELECT MAX(p2.Item_Price)
  6      FROM Product_Catalog p2
  7      WHERE p2.Cat_ID = p.Cat_ID
  8  );
```

CATEGORY_TITLE	PROD_NAME	HIGHEST_PRICE
Computing & Peripherals	Ultrabook Pro 15	69900
Activewear & Footwear	Trail Running Shoes	3600
Smart Home Living	Robotic Vacuum Cleaner	24000
Publications & Media	Clean Code Handbook	899

- Display categories having more than 5 products.

```
SQL> SELECT c.Category_Title
  2  FROM Product_Catalog p JOIN Item_Category c ON p.Cat_ID = c.Cat_ID
  3  GROUP BY c.Category_Title HAVING COUNT(p.Prod_ID) > 5;
```

CATEGORY_TITLE
Computing & Peripherals

- Find average product price category-wise.

```
SQL> SELECT c.Category_Title, AVG(p.Item_Price) AS Avg_Price
  2  FROM Product_Catalog p JOIN Item_Category c ON p.Cat_ID = c.Cat_ID
  3  GROUP BY c.Category_Title;
```

CATEGORY_TITLE	AVG_PRICE
Computing & Peripherals	37350
Activewear & Footwear	2725
Smart Home Living	20250
Publications & Media	899